

The complexity of compensation contracts

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Abstract

Management compensation is often categorized as either sensitive or insensitive to firm performance. This one-dimensional treatment ignores the variation in the types and terms of compensation contracts. Through a cross-sectional examination of shareholder-authorized compensation arrangements, this paper demonstrates that the terms of stock option and restricted stock plans, and the flexibility afforded the board of directors in negotiating with managers, vary systematically with the characteristics of the assets being managed. This variation in compensation contracting challenges theorists to incorporate the richness of management contracts into models of incentive pay.

Key words: Contracts; Executive compensation; Incentives

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1. Introduction

Despite considerable academic and media discussion of executive pay, our understanding of managerial compensation contracts is still incomplete. Private negotiations between management and the board of directors or the compensation committee generate agreements that are only partially observable to

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shareholders and outsiders, who must learn about managerial incentives from boiler-plate discussions in corporate proxy statements. To date, empirical researchers have relied on actual remuneration to draw inferences about these unobservable contracts. In this paper, I use publicly available information to investigate aspects of management compensation contracting on which previous research is silent.

The standard compensation scheme nested in theoretical agency models categorizes remuneration as either sensitive or insensitive to firm performance. For example, an option to purchase the firm's common stock at a fixed price is sensitive to firm performance whereas straight salary is not. This broad categorization of compensation ignores not only many important aspects of compensation contracts but also the flexibility available to the board of directors in negotiating with management.¹ For example, when an executive receives an option to purchase 100 shares of his employer's common stock, that right has many facets, including the exercise price, the terms of the option's exercise schedule, and the award of associated appreciation rights or restrictions on the disposition of the underlying stock. All are levers that, when adjusted, influence the sensitivity of pay to performance in ways not fully captured by the standard, one-dimensional treatment.

In this paper, I explore the various remuneration arrangements available to the board in negotiating managerial compensation contracts. Drawing on descriptions of shareholder-approved equity-awarding plans and multi-year cash incentive plans found in corporate proxy and registration materials, I report the *least restrictive* terms under which the board can contract. For example, two years of continued employment is a common prerequisite for the removal of resale restrictions on restricted stock awards. In administering a plan with such a prerequisite, a board can elect to offer restricted stock awards with waiting periods of greater than two years. However, it cannot remove resale restrictions earlier than the authorized two-year wait. This *ex ante* approach establishes bounds on unobserved compensation contracts. Analysis of these permissible contracting terms reveals considerable cross-sectional variation in the types and terms of contracts that boards can offer to managers.

This paper relies on a new data set compiled from information on the availability and terms of awards for stock option plans, stock appreciation rights plans, restricted stock arrangements, and long-term performance plans

¹ Throughout the paper, the board of directors is used to connote the shareholders' agent or representative in the process of negotiating executive remuneration. In fact, most large firms have a compensation committee composed of outside directors who work with senior management and compensation consultants to set the level and composition of executive pay. Many firms also have committees devoted solely to the administration of a single compensation plan, such as a stock option committee.

for 371 of the Fortune 500 firms for the year 1980. These data allow for cross-sectional analysis of the availability of individual arrangements, the bundling or combination of individual arrangements, the terms of authorized awards, and the degree of discretion available to the board in compensation contracting. While this investigation of contracting possibilities does not uncover the actual contracts negotiated between managers and the board, it does expose a richer contracting problem than is acknowledged in the theoretical literature.

This paper departs from studies that focus on individual compensation plans in isolation (e.g., Lambert, Larcker, and Larcker, 1989) and that implicitly assume that the decision to employ one type of compensation plan is unrelated to the use of another. In the spirit of Jensen and Murphy (1990), who recognize the interdependency of various forms of pay and incorporate cash, equity-based compensation, and the threat of dismissal in their analysis on the sensitivity of CEO pay to firm performance, the premise of my work is that different devices provide incentives. Consequently, it should not be surprising to observe variation across firms in the mechanisms used to provide incentives. In addition, most studies use *ex post* pay to examine managerial incentive contracts. My focus on the set of permissible contracts rather than the realization of a contingent contract complements the existing literature on executive compensation.

My *ex ante* approach to compensation contracting also highlights the tendency of *ex post* pay to understate contractual incentives. For example, suppose the board offers a manager the 'prize' of an outright grant of 50,000 shares of common stock if predetermined performance measures are met. If the manager fails to achieve these 'trigger' goals, his actual compensation will appear insensitive to firm performance. However, it is the invisibility rather than the absence of such unattained prizes or hidden incentives in *ex post* calculations that leads to an understatement of the relation between firm performance and executive pay. This example demonstrates how researchers lose valuable information by summarizing contingent rewards with a single realization, instead of taking an *ex ante* approach.

The next section develops the theories that predict cross-sectional variation in managerial compensation. Section 3 then describes the data and the sample selection process, and also defines specific compensation plans such as stock option, restricted stock, and long-term performance plans. Section 4 investigates the incidence and terms of these plans and also considers the bundling of equity-awarding plans. I examine the discretion or flexibility in compensation contracting vested in the board in Section 5. Section 6 concludes.

2. Theories predicting cross-sectional variation in incentive compensation

In crafting managerial compensation packages, the board seeks to reward prior contributions to firm performance and to motivate future profit-maximizing behavior. We observe an array of awards granted to executives, including

cash, stock, and options on stock. But what determines the forms of compensation used in a given firm?

2.1. Choice of performance measure

Standard principal–agent theory predicts that the ability of the board (here, the principal) to observe the manager's (agent's) input or effort influences the form of compensation. When appropriate actions are known and observable, the optimal incentive contract pays the manager a fixed salary and charges him for suboptimal behavior. In contrast, linking wealth to productive outcomes such as the value of the firm is necessary to induce the manager to behave optimally when managerial actions are unobservable (Hölmstrom, 1979).

Lambert and Larcker (1987) examine the theory in the context of cross-sectional variation in managerial compensation and ask whether the relative weight placed on a performance measure is an increasing function of the amount of information it conveys about the manager's actions. They observe that the relative use of accounting and market performance measures in executive compensation is affected by the amount of 'noise' inherent in the two signals and the 'sensitivity' of the signals to managerial actions. Specifically, they find greater use of stock-based compensation when the accounting measure is 'noisy' relative to the stock-based measure and when a firm is in the early stages of investment as characterized by rapid growth in assets and sales.

Thus, the information conveyed by performance measures about managerial actions should result in systematic variation in the choice of performance measures across firms. Managerial pay will not be tied to firm value when accounting data are a more reliable measure of the contribution of managerial effort to the value of the firm. If managerial decisions relate solely to the maintenance of existing assets, then the manager's ability to affect firm value is likely to be extremely limited. In such a setting, stock-based compensation will do a poor job of providing incentives, and alternative measures of performance, such as physical plant down-time, will provide a more accurate assessment of managerial effectiveness. In contrast, if managerial decisions have substantial value implications, as in the case of growth-option assets (Myers, 1977; Smith and Watts, 1992) or management buyouts (Kaplan, 1989), stock-based compensation will be effective in motivating value-maximizing behavior.

2.2. Contract length

Long-term contracts can encourage managers with specialized knowledge to remain with the firm. When a manager possesses valuable experience that is costly to transfer, his departure can impose large costs on the firm through a loss of efficiency and the potential for leakage of proprietary information. In an effort to retain managerial skill, executive compensation can be structured to bond

managers to the firm. In general, the use of deferred or forfeitable rewards raises the cost to the employee of leaving the firm and simultaneously increases the minimum price at which employees with specialized knowledge can be bid away.

Uncertainty about the appropriateness of managerial actions and the timing of the arrival of information that reduces this uncertainty are also relevant for determining the length of compensation contracts. Many actions are taken by managers, some of which have immediate implications for the value of the firm (e.g., shifting to a lower-cost supplier) and others for which there is persistent uncertainty about their impact on firm value (e.g., approval of a new line of basic research). Given that the benefits of managerial decisions of the latter type will not be immediately observable, motivating managers to make such decisions appropriately requires long-term contracting (Fudenberg, Hölmstrom, and Milgrom, 1990).

Variation in the payoff from bonding managerial capital to the firm, as well as uncertainty about the desirability of managerial actions and the length of time required to resolve that uncertainty, imply differences in the net benefit to the firm of long-term contracting. For projects with long gestation periods or requiring specialized knowledge, I expect to observe stronger restrictions on equity awards, such as longer waits to exercise stock option awards and more severe encumbrances on restricted stock grants. More generally, I expect to observe more frequent reliance on contractual terms that raise the cost to an executive of changing employers.

2.3. Agency problems with the board

The existence of agency problems between shareholders and the board can lead to suboptimal compensation contracts. The most likely scenario involves either a 'captured' or otherwise complacent board of directors that dutifully rubber-stamps the compensation-related proposals submitted by managers. The likely result is 'overpaid' executives for whom compensation is high relative to performance.

Using compensation plan data to detect management-dominated compensation negotiations is difficult. One manifestation of agency problems at the board pairs competitive salaries with low hurdles for incentive awards: by setting performance targets that are sufficiently low (i.e., return on assets below inflation) or setting exercise prices below market value at the time of grant, the board converts incentives into entitlements and pays above-market wages. (Authorization to grant in-the-money options is unlikely to appear in the data due to IRS stipulations for nonqualified tax treatment of option awards that require plans to award options at or above the 'fair market value of the stock on the date of award'.) Evidence of such a phenomenon requires both *ex ante* and *ex post* compensation data. Alternatively, discretion in determining the terms of awards,

while important for tailoring compensation contracts to particular circumstances, can be used by the board to indulge managerial preferences for easily obtained awards. This suggests that flexible compensation plans, with award terms determined at the discretion of the board, will be more likely in firms with insider-dominated boards than in other firms.

3. Data construction and sample selection

The compensation data used in this study are drawn from two classes of publicly available documents produced by corporate management and filed with the Securities and Exchange Commission (SEC): (i) annual proxy statements, Schedule 14a, and (ii) compensation plan registration statements, Form S-8. SEC rules require that management provide a discussion of existing managerial remuneration plans in the company's proxy materials. For purposes of proxy disclosure, a 'plan' is any agreement or arrangement by which executives are compensated, whether or not the plan itself is registered with the SEC. In practice, the discussions on compensation arrangements found in proxies range from one-line statements affirming the existence of a plan or employment agreement to detailed descriptions of incentive-compensation contracts, including performance objectives and reward schedules.

Information on compensation plans is also available as a by-product of institutional arrangements that require shareholder participation in the determination of stock-based pay. Federal securities laws require firms to receive the approval of a majority of its outstanding shares for the authorization of, and material amendments to, any remuneration plan that transfers equity to managers. For both stock option and employee stock purchase plans, the federal tax code requires that a proposed plan be submitted for shareholder approval within a year before or after its adoption to qualify for preferential tax treatment. In addition, the New York and American Stock Exchanges and the National Association of Securities Dealers, as well as several state incorporation laws, require listing companies to obtain shareholder approval for stock option plans.

To gain approval for a new plan or a change in an existing plan, management must present to shareholders a detailed description of the proposed plan. The additional information provided to shareholders in these cases that is not normally provided in routine proxy discussions includes the basis of eligibility for participation in the plan, the amount that would have been distributed under the plan had it been in effect during the firm's last fiscal year, and (in the case of an amendment to an existing plan) a summary of any significant differences from the existing arrangement. Embedded in this information are the terms under which shares can be granted to managers. Criteria for fully vesting shares, contingencies on the sale or transfer of awarded equity, and limitations on the

voting and/or dividend privileges associated with awarded shares are some of the terms that are finalized upon shareholder approval of a plan. Until recently, managers were required to hold equity purchases for a minimum period of six months. This constraint stemmed from Section 16(b) of the Securities Act of 1934 which prohibits corporate insiders from profiting from 'short-swing' transactions involving the matched purchase and sale of any of the firm's equity securities over a period of less than six months. The SEC has since adopted a broad exemption to this minimum holding period requirement for benefit plans, but acquisitions of stock per employment agreements or on the exercise of an option, warrant, or right were required to be retained by an insider for six months *after* receipt during the sample period.²

The contractual opportunities available to the board when negotiating with management can be summarized by the set of active compensation plans. I study the set of plans for which participation is limited to key, upper-level employees, omitting benefit plans such as savings, thrift, and stock purchase plans that are available to all employees of the firm or in which participation is voluntary. I concentrate on features over which the board has decision rights, such as the timing of resale restrictions, and ignore terms determined by the participant, such as the election of cash upon award or deferred cash. Because I do not include the actual remuneration of managers, I do not discuss the valuation of different forms of managerial compensation, nor do I make any attempt to quantify the intensity with which any given plan is employed in terms of either the number of executives participating in the plan or the size and frequency of awards.

3.1. Types of plans

Incentive compensation plans generally take the following forms: stock option, restricted stock, and long-term performance plans. Stock option plans award eligible participants the right to purchase a fixed number of shares of common stock at a predetermined exercise price over a finite horizon. The vesting of this right generally takes place over a number of years following the award date. A typical vesting schedule includes a minimum wait to exercise of 12 months, after which one-quarter of the award is available for exercise; the remainder of the award becomes available for exercise in equal installments over the next three years. The average wait to exercise for the typical plan is thus 30 months ($0.25(12 + 24 + 36 + 48)$). Prior to exercise, the value of an option

²Effective May 1, 1991, under Rule 16b-6(b), the granting of a stock option is a 'purchase' of the underlying stock for purposes of the short-swing profit rule. Therefore, it is no longer necessary for insiders to retain stock acquired by option exercise for six months *provided* at least six months elapse between receipt of the option and its exercise.

award is tied to the price of the firm's common stock. A manager can sever the link between an award and firm value by exchanging vested options, at the fixed exercise price, for unrestricted shares of common stock that can be sold subsequently for cash. In more than 65% of the 303 stock option plans in the sample, the board can award a stock appreciation right (SAR) together with an option grant. The tandem SAR enables a manager to earn the appreciation on the underlying stock *as an alternative* to purchasing and then liquidating the security. Thus, a tandem SAR costlessly converts an in-the-money option into cash equal to the market price of the stock less the option's exercise price. Typically, the departure of an executive from the firm requires him to forfeit unexercised options.

Restricted stock awards endow managers with a fixed quantity of shares of the firm's equity with restrictions on resale or transfer and a forfeiture clause that invalidates the award if the executive quits or is fired before the restriction period lapses. A minimum period of employment is required before restrictions lapse for some fraction of an award. The remainder of that award is 'earned out', or vested, in accordance with a predetermined schedule. Hence, prior to vesting and the sale of the stock, there is a direct link between firm value and managerial wealth. In general, the restricted equity arrangements observed in the sample convey all of the privileges of stock ownership, including dividends and voting rights, from the time of the award until the executive sells the shares. For this study, the shareholders of a firm are said to authorize the granting of restricted stock if either the firm has a formal restricted stock plan or the board is authorized to grant resale-restricted equity in accordance with any compensation arrangement (these conditions are met in 79 of the sample firms).

Finally, long-term performance plans set performance goals, typically in terms of accounting measures such as the growth in earnings per share over a specified horizon (normally ranging from three to five years). If performance objectives are met, earned awards are distributed in cash, equity, or a combination of cash and equity. Although performance-share and performance-unit plans are often lumped together as long-term compensation, the unit of account in which an award is carried over the performance period creates an important distinction between these two types of plans. In the case of performance-share plans, the award is accounted for in common-share equivalents, with the fraction of the award earned dependent upon the extent to which stated performance targets are met. Under this type of long-term plan, there are two factors that influence the size of the final award: (1) the degree to which the target is satisfied and (2) the change in stock prices over the performance period. Therefore, like equity awards, performance-share plans tie the recipient's reward to the appreciation of the firm's stock price over a specified period. In contrast, performance-unit plans grant fixed awards (e.g., 100 units each valued at \$75) that do not vary with the stock price. Consequently, the sole determinant of the final value of a performance-unit award is how closely the plan participants

come to achieving pre-specified performance targets. Among the sample firms, there are a total of 112 active long-term performance plans: 52 performance-share plans, 54 performance-unit plans, and six plans for which the unit of account cannot be determined. However, reporting on long-term performance plans is nonsystematic. Plans authorized to award equity to managers must receive shareholder approval and are therefore described in SEC filings, but performance-share and performance-unit plans that distribute cash awards exclusively need not be detailed in proxy statements, nor is management obligated to register the plan. Hence, the data on this type of plan, as garnered from 1979–1981 proxy statements, may be biased toward long-term performance plans that distribute stock awards.

3.2. *Data sample*

The sample of firms included in the analysis are the 371 of the 1980 Fortune 500 firms in Kole (1996). I exclude the other 129 firms because either the market-to-book ratio could not be calculated (85) or managerial ownership data are unavailable (44). In portions of the paper, I compare industry subgroups so as to exploit existing industry-level data on innovative activity. Each sample firm is classified according to its primary line of business based on the four-digit SIC code accounting for the largest percentage of sales as determined by Compustat: (1) food & kindred products (SIC 2000–2099, 37 firms), (2) chemicals & allied products (SIC 2800–2899, 53 firms), (3) metals, primary and fabricated (SIC 3300–3499, 39 firms), (4) machinery, industrial & electronic (SIC 3500–3699, 76 firms), (5) other manufacturing – consumer market (63 firms), and (6) other manufacturing – producer market (103 firms). The breakdown of the first four groups follows the categorization of manufacturing firms used by the U.S. Department of Commerce in the *Survey of Current Business*; the other two categories divide the remaining firms into relatively homogeneous groups. Matching these groups with an output-based measure of innovative activity (see Acs and Audretsch, 1990) produces the following ranking of the six subgroups from most to least innovative: machinery, 26.8 innovations per four-digit SIC code; chemicals, 21.8; producer, 19.2; metals, 9.9; consumer, 5.9; and food, 5.9.

To complement the categorization of firms by their innovative output, I subdivide the sample by expenditures on research and development (R&D) to create homogeneous groupings of firms determined by an input-based rather than an output-based measure of innovative activity. Three categories of research intensity are used: (1) firms with 1980 R&D expenditures of 1% or less of book value (175 firms), (2) firms with R&D expenditures between 1% and 5% of book value (138 firms), and (3) research-intensive firms that spent 5% or more of their book value on R&D in 1980 (56 firms). While the industry coverage is extremely broad in both the low- and middle-R&D groupings, the high-R&D

group consists of firms from the chemicals (SIC 28), machinery (SIC 35, 36), and equipment (SIC 37, 38) industries only.

Table 1 lists the variables used in this analysis, along with their definitions, units of measure, and sources. Plan data include four dummy variables (stock

Table 1
Variable definitions

Variable	Units	Definition (Sources) ^a
<i>Plan data:</i>		
Stock option	0, 1	Firm has a stock option plan in place.
Terms: Minimum wait to exercise	Months	Minimum grace period after which some portion of the stock option award may be converted to common stock.
Average wait to exercise	Months	Average of exercise restriction, assuming option is exercised as soon as restrictions lapse.
Stock appreciation rights (SARs)	0, 1	Firm either has a free-standing SAR plan or issues SARs in tandem with stock option awards.
Restricted stock	0, 1	Firm either has a formal restricted stock plan or has issued resale-restricted equity to top executives.
Terms: Minimum wait to vest	Months	Minimum wait until some portion of a restricted award is uncumbered or 'earned out'.
Long-term performance plan	0, 1	Firm has a performance plan that sets goals over a specified period of greater than one year.
Terms: Unit of account - stock	0, 1	Long-term plan accounts for award in common stock or stock-equivalents.
Unit of accounts - other	0, 1	Long-term plan accounts for award in units other than shares of common stock.
Stock payout	0, 1	Long-term plan authorizes the award of stock if performance objectives are met.
Cash payout	0, 1	Long-term plan authorizes the award of cash if performance objectives are met.
<i>Governance data:</i>		
Outsiders	%	Fraction of board seats held by nonemployee, unaffiliated directors (MacAvoy data, proxy statements).
Family	0, 1	Family representative either in management or on the board of directors (Schedule 14a, Dun's Reference Book of Corporate Management).
Stock ownership	%	Beneficial ownership of outstanding common stock held by officers and directors as a group (Corporate Data Exchange volumes).

Table 1 (continued)

Variable	Units	Definition (Sources) ^a
<i>Firm characteristic data:</i>		
Book value of assets	\$B	Inflation-adjusted book value of physical plant, inventories, and investments in unconsolidated subsidiaries in billions of 1980 dollars (Compustat).
R&D	%	Research and development expenditure (1980) per book value dollar of assets (Compustat).
Market-to-book	%	Ratio of firm's market value, calculated using the average of the high and low common stock price for 1980, to the book value of physical assets (NBER Griliches R&D tape).
Lines of business	Number	Count of four-digit SIC codes in which the company operated in 1980 (Standard and Poor's Corporate Register).

^a Plan data is extracted from corporate proxy and registration statements.

option, stock appreciation rights, restricted stock, and long-term performance) that equal one if the firm has an active plan in place in 1980 and zero otherwise. A plan is active if it has securities and/or dollars available for distribution and has not expired. Individual plan data are aggregated into a dummy variable (equity-authorizing) equal to one if the firm has at least one remuneration plan that authorizes the award of stock. I omit annual performance plans from the analysis both because there is little variation in the sample (over 95% of the firms studied have annual bonus or incentive arrangement) and because management rarely provides information about the target or objective that triggers payoff in annual plans. I also omit individualized employment agreements.

4. Boundaries on compensation contracting

In this section, I test the predictions from Section 2 regarding cross-sectional variation in compensation plans. The principal-agent literature predicts that when appropriate managerial actions are known but unobservable, linking pay to measurable performance is necessary to induce managers to behave optimally. To proxy for the inability to observe managerial actions, I use firm-level annual expenditures on research and development as a fraction of the book value of assets on the theory that a substantial investment in R&D projects makes determination of managerial value-added more difficult and thus leads to greater reliance on stock-based pay. Alternatively, if R&D activity can substantially influence firm value, greater reliance on stock-based pay is predicted in

research-intensive firms. The ratio of the market value to the book value of the firm's assets proxies for the difficulty in measuring managerial performance in the presence of intangible assets. Together, these proxies allow a test of whether the sensitivity of firm value to managerial effectiveness influences firms' choices in compensation plans.

The composition of the board should also affect the availability of compensation plans. Compensation plans can provide a mechanism through which wealth is transferred from shareholders to managers. If insiders control the board, they will seek shareholder approval of plans that enable the board to deliver stock and/or cash payments to managers. While these payments can impose risk on managers, disclosure of remuneration from compensation plans is more easily buried in proxy statements than are bloated salaries. Thus the theory predicts that, *ceteris paribus*, individual plans are more likely to be observed when there is a smaller representation of outsiders on the board. However, casting the board as an exogenous factor driving compensation contracts is not without complications. For example, if management seeks an amicable board to certify its preferred compensation arrangements, then the slate of directors presented to shareholders will reflect this agenda much in the same way U.S. Supreme Court nominees reflect the agenda of the nation's president (see Hermalin and Weisbach, 1996). Thus, I interpret the following multivariate test results as providing preliminary, rather than definitive, evidence on the agency problem hypothesis.

The tests on the availability of component plans also include a proxy for firm size as well as two additional governance measures: a dummy variable that equals one if a representative of a founding family fills a top management position or board seat, and the fraction of outstanding common stock held by officers and directors as a group. Descriptive analyses show that large firms tend to pioneer the use of new forms of compensation (Fox, 1983, 1984). To the extent that restricted stock plans were less prevalent in 1980 than other equity-based plans, firm size may provide an explanation. The ownership variables test whether the ownership of equity by top management influences the contracting set for all managers. Univariate tests reveal that formal compensation plans are rare among family firms; whether this result is robust in multivariate tests is of interest.

Table 2 reports the estimated probability derivatives from logit equations predicting the availability of executive compensation plans for the sample of 371 of the 1980 Fortune 500 firms. For example, the results on marginal effects reported in column 1 show that a 1% increase in the ratio of R&D to book value increases by 0.8% the likelihood that a firm will have at least one compensation plan that authorizes equity remuneration for managers. Asymptotic *t*-statistics are reported in parentheses below the estimated probability derivatives. Each column provides an estimate of how the variables that proxy for the composition of the firm's assets (market-to-book value of assets, R&D-to-book value of assets), measures of governance (officer and director stockholdings, the

Table 2

Estimated probability derivatives from logit equations predicting the availability of executive compensation plans for a sample of 371 of the 1980 Fortune 500 firms^a

	(1)	(2)	(3)	(4)	(5)
	Equity- authorizing plan ^b	Component plans		Performance ^c	
		Stock option ^d	Restricted stock ^d	Share	Unit
Intercept	-0.01 (-0.13)	-0.01 (-0.08)	-0.46 (-2.90)	-0.35 (-2.69)	-0.32 (-2.36)
Market-to-book value of assets	0.06 (2.11)	0.11 (2.31)	-0.00 (-0.03)	0.01 (0.21)	-0.01 (-0.32)
R&D book value of assets	0.76 (1.22)	0.82 (0.91)	1.48 (2.03)	-1.22 (-1.57)	0.01 (0.01)
Family ^e	-0.07 (-3.21)	-0.12 (-2.69)	-0.02 (-0.37)	-0.10 (-1.59)	-0.08 (-1.21)
Percentage stockholdings of officers and directors ^f	-0.05 (-0.91)	-0.10 (-0.86)	-0.11 (-0.63)	-0.02 (-0.17)	-0.06 (-0.41)
Percentage of outside board members ^g	0.05 (1.01)	0.08 (0.78)	-0.15 (-1.28)	0.02 (0.23)	0.04 (0.43)
Logarithm of the book value of assets	0.01 (1.08)	0.02 (0.92)	0.04 (2.28)	0.02 (1.61)	0.02 (1.01)
Likelihood ratio statistic ^h	40.04 [0.0001]	30.10 [0.0001]	14.17 [0.023]	10.43 [0.108]	5.13 [0.505]
Number of firms with active plan(s) in 1980	336	303	79	52	54

^aThe numbers reported in the table are $\partial \text{Prob}(\text{Plan}) / \partial x_i$ for each variable, evaluating all independent variables at their means. Asymptotic *t*-statistics are represented in parentheses.

^bDependent variable equals one if the firm has a compensation plan that authorizes the award of stock.

^cDependent variable equals one if firm has a stock option plan.

^dDependent variable equals one if firm has a compensation plan that authorizes grants of stock that cannot be sold for a fixed period of time.

^eDependent variable equals one if (share) firm has a long-term performance plan that accounts for the award in shares of common stock or its equivalents or if (unit) firm has a long-term performance plan that accounts for the award in units.

^fVariable equals one if there is a family representative either in management or on the board of directors.

^gBeneficial ownership of outstanding common stock for officers and directors as a group.

^hFraction of board members who are neither firm employees (current or retired) nor affiliated with the firm.

ⁱThe likelihood ratio test is computed to test the hypothesis that all the parameters in the model are simultaneously equal to zero. Under the null hypothesis, the test statistic has an asymptotic distribution that is chi-square with the degrees of freedom equaling the number of parameters in the model. The values in brackets denote the *p*-value of the statistic.

percentage of all board members who are neither employed by nor affiliated with the firm, the participation of the founding family in corporate management), and firm size influence the likelihood that a specific compensation plan will be in place.

The results reported in columns 1–3 of Table 2 reveal that the availability of any plan that authorizes stock awards increases with the two proxies for the relation between firm value and managerial effectiveness (R&D-to-book and market-to-book). The results on stock option plans (column 2) support the predictions of Smith and Watts (1992) and affirm the results of Gaver and Gaver (1993). The estimated influences of the two proxies vary by type of plan, with the market-to-book measure being a significant predictor of stock option plans and R&D-to-book predicting the incidence of restricted stock arrangements. However, given the correlation between the market-to-book and R&D variables, elimination of either proxy increases the significance of the estimate of the remaining proxy to conventional levels. Interestingly, while these proxies for the firm's asset composition do not explain the use of performance-unit plans, there is weak evidence that the incidence of performance-share plans *decreases* with firm-level expenditure on R&D (p -value of 0.1157.) This suggests that in research-intensive firms, pure stock-based compensation plans dominate the combination of accounting- and stock-based remuneration that is characteristic of performance-share plans. This is not surprising since accounting data are historical in orientation and thus do a poor job attributing value to investments with future revenue inflows. This finding is consistent with the assertion in Smith and Watts (1992, p. 276) that accounting numbers are 'poorer measures of performance' when the assets being managed are growth-option (high market-to-book value) assets. It should be noted that the results on firm characteristics are not driven by industry effects. The inclusion of industry dummies does not alter the qualitative results on the availability of individual plans, suggesting that my findings are robust both within and across industry groups.

Controlling for firm size, participation in corporate management by a founding family significantly reduces the likelihood of establishing most types of equity-awarding plans. This finding is consistent with at least two interpretations: (1) due to the already high level of managerial stock ownership, there is less reliance on equity compensation in family firms (in this sample, the mean ownership stake held by officers and directors as a group is 24.2% and 8.1% for firms with and without founding family representation among top management, respectively); and (2) in general, there is less formal or explicit contracting in family firms. The estimated negative, though statistically weak, influence of family participation on the incidence of purely accounting-based long-term performance plans lends support for the second explanation.

Contrary to Lambert and Larcker's (1987) finding that the relative weight placed on a market or stock-based performance measure is a decreasing function of management's ownership of the firm, the results in Table 2 show that the

level of managerial ownership has little effect on the availability of plans. This suggests that beyond the typically large equity share held by founding families, the ownership of the current management team does not influence *ex ante* compensation contracting.

Finally, the fraction of board seats held by outsiders does not significantly influence the likelihood of firms' adoption of any of the incentive plans under investigation, suggesting that the composition of the board does not alter the menu of plans available to the board for compensation contracting. This interpretation, however, rests on the assumption that the composition of the board is exogenous or predetermined.

4.1. Approving multiple equity-based plans

This section combines data on the availability of stock option and restricted stock plans to investigate the complementary nature of these contracting devices. If boards propose, and shareholders subsequently ratify, equity plans that link managerial rewards to firm value, then that link can be created using either stock options or restricted stock. For increases in the value of the underlying stock above the option exercise price, restricted stock and stock options deliver the same *ex post* payout. However, both because stock options may be out-of-the-money and because the terms of restricted stock and stock option grants can differ, restricted stock and stock options can play different roles in compensation contracts. For instance, to receive preferential tax treatment, the life of a stock option award cannot exceed ten years, while no such upper bound exists on the maturity of restricted stock grants. This limits the applicability of stock options in long-term contracts. Thus, in firms for which the contracting horizon can exceed ten years, it is optimal to include both stock options and restricted stock in the menu of compensation plans from which managerial contracts are written.

A second explanation for offering both restricted stock arrangements and stock option plans also relates to their complementary nature (Milgrom and Roberts, 1990). If increasing the level of one form of compensation raises the marginal return to increases in another form of pay, then the two types of rewards are said to exhibit 'complementarity'. For example, if bonding a manager to the firm through restricted stock reduces his willingness to take desirable risks (such as risk associated with positive net present value investments), then there are incremental gains to the firm from granting options concurrently with restricted stock (assuming, of course, that the option award's inducement to bear risk exceeds the disincentive created by increasing the manager's exposure to firm-specific risk). As argued in Section 2.2, bonding a manager to the firm through restricted stock is optimal when there is persistent uncertainty about the value implications of investment decisions or when a key decision-maker has valuable knowledge that is costly to transfer.

Table 3 investigates the issue of whether shareholders authorize stock option awards and restricted stock grants concurrently. The sample is subdivided to highlight high-R&D firms for which longer-term compensation contracting is predicted. The cells contain the proportion of research-intensive firms (firms with $R\&D/Book \geq 5\%$) and all other firms (those firms with $R\&D/Book < 5\%$) that have a stock option plan, a restricted stock plan, both plans, or neither plan. As indicated by the Z-statistic reported in each cell, research-intensive firms are significantly less likely to have neither plan (3.6% compared to 15.2%) and significantly more likely to adopt both types of equity-awarding plans (37.5% compared to 12.7%). As predicted by complementarity and long-term contracting considerations, restricted stock and stock option plans are simultaneously authorized by shareholders of research-intensive firms. Of the 315 firms with an R&D-to-book ratio below 5%, roughly three-quarters employ either a stock option or a restricted stock plan. Indeed, the simple correlation between the incidence of restricted stock and stock option plans for these 315 firms is negative and significantly different from zero at the 6% level.

Table 3

The proportion of firms with stock option plans, restricted stock arrangements, or both, by research intensity for 371 of the 1980 Fortune 500 firms (number of firms in brackets)

Stock option plan	Restricted stock arrangements	
	No plan	Active plan
No plan		
R&D book < 5% (N = 315)	15.2% [48]	5.4% [17]
R&D book $\geq 5\%$ (N = 56)	3.6 [2]	1.8 [1]
Z_c^* (z-value)	2.36 (0.0091)	1.16 (0.1230)
Active plan		
R&D book < 5% (N = 315)	66.7% [210]	12.7% [40]
R&D book $\geq 5\%$ (N = 56)	57.1 [32]	37.5 [21]
Z_c^* (z-value)	1.38 (0.0838)	4.61 (0.0001)

*The Z_c statistic is calculated to test the difference between two population proportions.

4.2. Implementation of individual plans

4.2.1. Stock option plans

Of the numerous terms of an option award that might vary across plans, there is considerable uniformity in four separate terms: (1) the award's exercise price is set at 100% of the fair market value of the firm's common stock on the date of award in 93% of the plans for which this information is available;³ (2) the sources for all shares sold to employees in every plan observed in the sample is authorized but unissued Treasury shares; (3) all but four (1%) of the sample firms offer nonqualified options that expire ten years after grant; and (4) all option awards granted pursuant to the plan are nontransferable. The construction of stock option plans to comply with the Internal Revenue Code and with Section 16 of the Securities Exchange Act probably explains the lack of cross-sectional variation in these terms. However, cross-sectional variation is observed in the linking of managerial wealth with firm value, i.e., in the exercisability of awards.

Columns 3 and 4 of Table 4 contain the minimum and average waiting periods, respectively, after which an option award can be exercised as per the shareholder-approved plan. Across the R&D subgroups, both the minimum wait to exercise any fraction of an option award and the average wait, calculated by assuming that all options are exercised when the award vests, are highest among the research-intensive firms. Although these results are only significant at the 10–15% significance level for a one-tailed test, the means of the minimum and average waiting periods omit firms with option plans for which the waiting periods are open or at the discretion of the Board (reported in columns 5 and 6 of Table 4). These discretionary plans are most prevalent among high-R&D firms. If the flexibility afforded to the board by an open or discretionary term is used most often to elongate waiting periods, as predicted when there is uncertainty about the timing of existing projects' payoffs, then the weak tendency toward longer waiting periods in the option plans of research-intensive firms may underestimate the severity of restrictions among these firms. If, alternatively, discretionary terms are interpreted as giving the recipients their most preferred terms (e.g., a 'blank check' for managers negotiating with a captured board), then the mean wait-to-exercise time summarized in columns 3 and 4 may overstate the waiting periods. Section 5 considers the issue of flexibility in contracting.

For the industry subgroups, where the availability of option plans does not differ significantly, the chemicals, machinery, and producer firms have higher

³ Of the remaining 7%, all but one plan authorizes in-the-money options. Gulf & Western Industries' 1973 Key Employee Stock Purchase Plan provides for option awards to its CEO and COO with embedded exercise prices equal to not less than 105% of the fair market value of the firm's common stock.

Table 4

Details of pre-approved stock option plans, by subgroup: availability of option plans, fraction of plans with tandem stock appreciation rights (SARs), and minimum and average waiting periods (in months) on awards as per the plan

Subgroup	(1) Active option plans [number] ^a	(2) % in (1) that authorise tandem SARs	(3) Wait to exercise (months) Minimum [number] ^a	(4) Average [number] ^a	(5) % firms in (1) for which (3) is open ^b	(6) % firms (1) for which (4) is open ^b
Full sample	81.7% [303]	67.0%	13.5 [262]	23.6 [217]	12.5%	26.1%
Low R&D R&D book $\leq 1\%$	77.1% [135]	64.4%	13.7 [119]	22.8 [101]	10.4%	21.5%
Middle R&D 1% < R&D book $\leq 5\%$	82.1 [115]	73.0	12.5 [101]	23.0 [83]	12.2	27.0
High R&D R&D book $\geq 5\%$	94.6 [53]	60.4	15.6 [42]	27.8 [33]	18.9	35.8
K-W test ^c χ^2_3 Pr $> \chi^2_3$	8.69 0.0130	3.34 0.1884	3.49 0.1748	2.52 0.2842	2.52 0.2837	4.14 0.1264
Chemicals	83.0% [44]	72.7%	14.7 [38]	25.2 [33]	13.6%	25.0%
Machinery	88.2 [67]	65.7	13.4 [54]	27.5 [46]	17.9	29.9
Producer	75.7 [78]	70.5	15.9 [67]	27.5 [52]	12.8	28.2
Metals	79.5 [31]	58.1	11.6 [27]	18.5 [22]	12.9	29.0
Consumer	84.1 [53]	67.9	12.9 [48]	21.2 [44]	7.5	13.2
Food	81.1 [30]	60.0	9.4 [28]	12.9 [20]	6.7	33.3
K-W test ^c χ^2_3 Pr $> \chi^2_3$	5.00 0.4154	2.94 0.7098	15.23 0.0094	24.89 0.0001	3.95 0.5560	6.20 0.2874

^aNonparametric test that the distribution of the variable has the same mean across different groups. This is a nonparametric analogue to a one-way analysis of variance.

^bTerm is determined at the discretion of the board of directors or the compensation committee.

^cTotal number of observations used to calculate the associated summary statistic.

mean minimum and average waits to exercise option awards. These intergroup differences are relatively strong and show that exercise terms are more restrictive for firms in industries with relatively high innovation rates. If high innovation rates are correlated with specialized knowledge, these results support the hypothesis that the restrictions on equity awards will be more severe in firms for which specialized knowledge is an important asset. The results are also consistent with the hypothesis that the timing of information revelation affects contract length if the average duration of investment projects on which managers are evaluated is longer in firms with more innovative activity.

The incidence of option plans across R&D subsamples, reported in Table 4, shows a strong positive correlation between the existence of option plans and firms' research intensity. Interestingly, the availability of tandem SARs, which enable a manager to receive the difference between the market price of the common stock and the predetermined exercise price without purchasing the underlying equity, is *lowest* among the high-R&D firms (see column 2). This suggests that although stock option plans are most prevalent among research-intensive firms, the plans in place in these firms are least likely to enable participants to sever the link between the award and firm value.

A handful of firms require managers to retain stock acquired by option exercise. Amsted Industries and Ingersoll Rand require managers to hold these shares for three years, while Abbott Laboratories, Harsco, Kellwood, and Tosco require managers to retain acquired shares for the remainder of their tenure with the firm. These additional restrictions prolong the incentive alignment effect of an otherwise identical option award. Such long-term contracts are predicted in firms in which managers have valuable specialized knowledge and/or for which long-term investment decisions involve uncertainty, and, in fact, each of the firms employing these forced stock-ownership arrangements outspent its industry average on annual R&D as a fraction of the book value of firm assets and had a market-to-book value above the mean for its industry. While some firms use an 'investment not resale' clause in stock option plans as a means of avoiding SEC registration and its associated fees, this institutional explanation does not account for these six cases: S-8 registration forms for the six stock option plans are on file with the SEC.

4.2.2. Restricted stock arrangements

Subsample analysis in Table 5 shows that the incidence of restricted stock arrangements is highest among research-intensive firms and firms in innovative industries (chemicals, machinery, and producer). Whereas only 21% of the sample firms have plans that authorize granting resale-restricted equity to managers, nearly 40% of the boards of the 56 high-R&D firms have this authority. Similarly, the incidence of the authority to grant restricted stock is significantly higher among industries that are characterized by high levels of innovative activity.

Table 5

Details of pre-approved restricted stock plans, by subgroup: the availability and minimum vesting period (in months) on awards as per the plan

Subgroup	(1) Active restricted stock plans [number] ^a	(2) Want to vest (in months) Minimum [number] ^a	(3) % firms in (1) for which (2) is date of retirement	(4) % firms in (1) for which (2) is at board's discretion
Full sample	21.3% [79]	28.7 [46]	27.8%	7.6%
Low R&D R&D book $\leq 1\%$	17.7% [31]	32.7 [18]	25.8%	6.5%
Middle R&D $1\% < \text{R\&D book} < 5\%$	18.6 [26]	32.8 [17]	19.2	7.7
High R&D R&D book $\geq 5\%$	39.3 [22]	15.8 [11]	40.9	9.1
K-W test [*]				
χ^2	12.74	5.59	2.86	0.13
Pr $> \chi^2$	0.0017	0.0612	0.2397	0.9387
Chemicals	28.3% [15]	23.0 [6]	40.0%	13.3%
Machinery	27.6 [21]	30.0 [12]	33.3	4.8
Producer	26.2 [27]	28.7 [19]	25.9	3.7
Metals	12.8 [5]	0 [1]	20.0	20.0
Consumer	9.5 [6]	27.0 [4]	0	16.7
Food	13.5 [5]	42.0 [4]	20.0	0
K-W test [*]				
χ^2	13.04	5.35	4.04	3.69
Pr $> \chi^2$	0.0230	0.3751	0.5441	0.5948

^{*}Nonparametric test that the distribution of the variable has the same mean across different groups. This is a nonparametric analogue to a one-way analysis of variance.

^aTotal number of observations used to calculate the associated summary statistic.

While the concentration of restricted stock plans is highest among high-R&D firms, the data suggest that the mean minimum vesting period is *shortest* for these firms. However, valuable qualitative information is lost in constructing a quantitative measure of the minimum transfer-restriction period. In addition

to the few cases in which the vesting schedule is unavailable or is determined at the time of the award, nearly 30% of the observed arrangements specify a minimum holding period equal to the recipient's date of retirement or beyond. As shown in column 3 of Table 5, these hold-until-retirement plans are concentrated in the high-R&D subsample: 9 (41%) of 22 restricted stock plans observed among research-intensive firms prohibit the disposition of restricted stock awards until retirement, compared with only 13 (23%) of the 57 remaining firms with active restricted stock arrangements.

To provide a more comprehensive characterization of vesting periods for restricted stock plans in the sample, I incorporate CEO age data and an assumption about age at retirement. The mean (median) age of the nine CEOs of high-R&D firms with hold-until-retirement restricted stock arrangements is 54 (54) years, compared with 61.5 (62) years for the remaining 13 CEOs whose firms have active restricted stock plans. To estimate the length of the minimum wait to vest for awards that are restricted until retirement, I calculate the difference between the age of the CEO in 1980 and an assumed retirement age of 65. When the age of the CEO is greater than 65 in 1980, I record a zero for the minimum wait to vest (there are three instances of this).⁴ Using this approximation, the minimum wait to vest is significantly longer in hold-until-retirement plans among high-R&D firms than it is for other firms with similar vesting schedules: the effective length of the hold-until-retirement restriction is 137 months or more than 11 years for the nine high-R&D firms' plans compared to 37 months for the 13 remaining hold-until-retirement restricted stock arrangements (*t*-statistic = 3.44). Incorporating these estimates into the calculations of the minimum wait to vest for all restricted stock plans *reverses* the original finding: using both actual CEO ages and plan-authorized vesting schedules, the minimum length of time a restricted stock award must be held by a manager before resale restrictions lapse is 74 months, on average, for plans found among high-R&D firms and 34 months for plans found among the low- and middle-R&D firms. This difference is significant at the 6% level. Disaggregating the low- and middle-R&D observations yields a minimum wait to vest of 36 months for the 27 low-R&D firms and an average minimum wait of 31 months for the 21 middle-R&D firms. An *F*-test on the difference between these means yields $F_{2,66} = 4.06$ (Prob > *F*: 0.02).

The effective minimum wait to vest for restricted stock arrangements varies significantly across industry groups. Collectively, the restricted stock contracts are more restrictive in innovative firms: chemicals, machinery, and producer manufacturing have a mean wait to vest of 50 months while executives in the

⁴Using data provided by Robert Parrino on CEO turnover for 177 CEOs of manufacturing firms, I conclude that there is no significant difference in the CEO's age at retirement by either industry or research intensity.

metals, foods, and consumer manufacturing firms wait only 20 months, on average, for resale restrictions to lapse on restricted stock grants. This difference is statistically significant at the 2% level.

The Motorola Executive Incentive Plan (MEIP) is an interesting restricted stock arrangement that effectively imposes career-long investment in the firm on employees who receive large cash incentive awards. In the event that an employee earns a MEIP award in any one year that exceeds 50% of base salary, that employee must use the amount in excess of 40% of base salary, less applicable income taxes, to purchase Motorola common stock following the exercise of a company-awarded stock option. This conversion from option to directly held stock must be made no later than 12 months after the date of the MEIP award. In addition,

such employee must advise the Compensation Committee that it is such employee's intention and commitment to the Compensation Committee to acquire such stock as a permanent investment in the Company; and no Equity Award may be granted to an employee who has failed to comply with the spirit and intent of such commitment and intention with respect to an Equity Award made to such employee in a prior year. [Motorola's 1981 Proxy Statement, pp. 15–16]

Given the high frequency of 'hold until retirement' restrictions among high-R&D firms, it is not surprising that corporate policy dictates that shares be acquired for 'investment not resale' at R&D-driven Motorola. In his 1980 Letter to Stockholders, CEO Robert W. Galvin emphasized the importance of 'increasing investment in key *long-term* strategic opportunities' (*italics mine*) to ensure a prosperous future [Motorola's 1980 Annual Report, pp. 3–4]

4.2.3. *Long-term performance plans*

Owing to the nonsystematic disclosure of long-term performance plans discussed earlier, analysis of the terms of this type of plan is limited. For the sample of long-term plans as a whole, the interval over which performance is measured ranges from two to six years in length, with the median performance period equal to four years. The length of the performance period is similar across performance-share and performance-unit plans, as is the variation in the use of accounting-based performance objectives or goals in the two types of performance plans. Roughly three-quarters of the performance goals or targets found in either the 44 performance-unit plans or the 31 performance-share plans for which this information is available relate to earnings growth over the performance period. Of these 75 cases, the payout of the share or unit award is solely contingent on growth in earnings per share in 46 cases (61%). An additional 11 plans (15% of the observed targets) combine an earnings-per-share measure with a second performance objective, such as the average return on total capital or a relative performance ranking among a set of designated firms.

5. Flexibility in the contracting process

The obligation on the part of the board to grant awards that conform to the terms of shareholder-approved plans limits the flexibility of the board in contracting with management. In this section, I explore the cross-sectional variation in the authorization of discretion by shareholders. I find that the degree of discretion or contractual flexibility placed in the hands of the shareholders' agent varies with the characteristics of the assets being managed in that firms that are larger, more diversified, and more research-intensive tend to build flexibility into the process of contracting for management pay.

There are at least three aspects of the process of compensation contracting that allow for the exercise of judgment by the board. First, although there are guidelines in most remuneration plans that direct the board to identify and reward key decision-makers, the assessment of who is 'key' and how much to award each 'key' employee is ultimately made by the board. Second, within a given menu of approved remuneration plans, the actual composition of the compensation package is decided by the board. Third, at the time of grant the board holds decision rights over the actual terms of equity awards. While most shareholder-approved stock plans predetermine the terms of equity awards, approximately one-quarter of the firms considered in this study explicitly authorize their negotiating agent (the board) to fix the restrictions on equity awards at the time of the grant.

Whereas flexibility in the contracting process creates the potential in all firms for the expropriation of shareholder wealth, the benefits of flexibility vary systematically across firms. In this sample, I observe that the life of equity plans ranges from three to five years, at which time a plan is either replenished with additional shares, substantively amended, or terminated. This timing appears to be related to frequent revisions in the tax treatment of stock-based compensation instruments during the 1970's and into the early 1980's. Given this periodicity, I expect that there is greater value to flexible contracting when the process being managed is evolving rather than static. For example, in research-intensive firms whose new products are continuously evolving from developmental to market stages, the ability to tailor executive compensation awards to coincide with the expected horizon of product development is beneficial. In contrast, the benefits of flexibility in determining the terms of stock awards are less obvious for more routine businesses for which there is little variation over time in product market activities.

Diversification provides another possible explanation for the formalization of discretion in compensation plans. Differences across operating units in the appropriateness of performance measures and length of contract suggest intrafirm variation in the compensation contracts negotiated with individual managers. Just as uncertainty regarding the future activities of the firm increases the benefits of contractual flexibility, a versatile menu of compensation alternatives complements line-of-business diversity.

Agency problems with the board provide a third explanation for the formalization of discretion in managerial compensation contracting. Under this hypothesis, giving discretion to structure compensation contracts provides a blank check with which a captured board can indulge managerial preferences. The outcome is suboptimal compensation plans that are slipped past unwitting shareholders.

To test for the hypotheses related to flexibility in contracting, I use logit models to predict the explicit provision of discretionary terms in executive compensation plans. Table 6 reports the results. The dependent variable is equal to one if some or all of the vesting schedule of an option or restricted stock award or the length of the performance period in a long-term plan is determined at the discretion of the board. Consistent with the hypothesis regarding uncertainty about future activities of the firm, the explicit provision of contractual flexibility is most likely to be found in high market-to-book, research-intensive firms. Greater product-line diversity is also shown to increase the likelihood of

Table 6

Firm equations predicting the provision of discretionary terms in any of the firm's executive compensation plans for 371 of the 1980 Fortune 500 firms^a

Independent variable	(1)	(2)	(3)
Intercept	-1.81*** (-4.40)	-1.88*** (-4.56)	-1.54** (-4.21)
Market book	0.32 (1.49)	0.52*** (2.86)	
R&D book	7.03 (1.61)		10.82*** (3.04)
Percentage of outside board members	-0.36 (-0.53)	-0.19 (-0.28)	-0.51 (-0.77)
Number of four-digit SIC codes in which the company operates	0.04** (2.25)	0.04** (2.19)	0.04** (2.17)
Likelihood ratio statistic ^b	15.04 0.0046	12.47 0.0059	12.68 0.0053

^aDiscretion is equal to one ($N = 88$) if any of the following terms are determined at the discretion of the board of directors or the compensation committee: (i) the exercise price of the option award ($N = 1$); (ii) some or all of the vesting schedule in either the firm's stock option ($N = 81$) or restricted stock plans ($N = 7$); or (iii) the length of the interval over which performance is measured in the firm's long-term performance plan ($N = 4$). Asymptotic t -statistics are represented in parentheses.

^bThe likelihood ratio test is computed to test the hypothesis that all the parameters in the model are simultaneously equal to zero. Under the null hypothesis, the test statistic has an asymptotic distribution that is chi-square with the degrees of freedom equaling the number of parameters in the model. The value in brackets denotes the p -value of the statistic.

Significant at the 1% (***), 5% (**), 10% (*) levels.

explicit contractual flexibility: the more four-digit SIC codes in which the firm operates, the higher is the likelihood that 'at the discretion of the board' clauses will be found in compensation plans. Similar results are found when three-digit SIC codes are used to proxy for product-line diversity. Though negative, the statistically insignificant coefficient on the fraction of outsiders on the board provides little support for the suboptimal contracting predicted by agency problems between shareholders and the board. However, this interpretation rests on the presumption that the composition of the board and the terms of executive compensation plans are determined independently.

A second test on contractual flexibility not reported here reveals consistent findings. I investigate which factors predict the number of equity-alternatives available to the board, within the sample range of zero to three plans, and find that multiple means of delivering equity to managers are available to the boards of firms with greater size, research intensity, and product-line diversity.

6. Conclusions

This paper considers the use and details of different remuneration devices in an attempt to deepen our understanding of the ways in which cash and equity-based awards are used to provide incentives to managers. The cross-sectional evidence on compensation contracting reveals that the authority of the board to grant equity can be predicted by the mix of tangible and intangible assets in the firm and, to a lesser extent, by firm size and founding family participation in management. In addition to systematic variation in the adoption of individual plans by firms, the terms of seemingly similar compensation vehicles such as stock options and restricted stock grants vary cross-sectionally. In particular, the board of a research-intensive firm is constrained to grant awards that are more restrictive (e.g., awards with longer time to full vesting) than is the board of a less innovative firm. Moreover, innovative firms are more likely to use a wider variety of equity-based awards. Also, the boards of firms that are larger, more diverse, and more research-intensive are more likely to have greater flexibility in contracting. This evidence suggests that the standard characterization of compensation as either sensitive to firm performance or performance-insensitive is overly simplistic: shareholders position the board to fine-tune compensation contracts along multiple dimensions to amplify or repress managerial incentives.

Given the existence of cross-sectional variation in *ex ante* compensation, a necessary next step to further our understanding of whether or not managers are being 'properly' compensated is to incorporate the variation in management compensation contracting into firm-level analyses of executive pay. To the extent that firm-specific menus of compensation arrangements are optimally

configured, the heterogeneity documented in this paper suggests that uniformity in contracts, and ultimately in pay, is unlikely.

The empirical evidence presented in this paper poses some interesting issues for future research. First, I find that the presence of a representative of the founding family on the board reduces the likelihood of any form of explicit compensation arrangement. This raises the question as to the form of incentive contracts for hired managers in publicly traded family firms. Second, an important dimension in which compensation plans can vary is in the terms of award forfeiture in the event that the executive leaves the firm prior to full vesting. Investigation of these terms should further our understanding of how firms contract out of the 'final period' problem. Finally, the observed multidimensionality of stock-based compensation awards raises questions about the appropriateness of a one-dimensional treatment of compensation in theoretical models. Future models of incentive pay should incorporate specific provisions of management contracts to determine their effect on firm performance.

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